Varsini Sakthivadivel Ramasamy

Baltimore, Maryland
varsini-sakthi.github.io | github | 443-802-2041

EDUCATION

Johns Hopkins University – Baltimore, MD *May 2026*
Master of Science in Bioinformatics (3.9/4.0)
Thesis: scRNA-seq Representation Learning for Cell-Type Classification

VIT University – Vellore, India *May 2024*
Bachelor of Technology in Biotechnology (3.9/4.0)

RESEARCH EXPERIENCE

Research Assistant, *Johns Hopkins University School of Medicine*, Baltimore, MD *Jan. 2026 – Present*
PI: Dr. Maide Ozen

Project: SpatialFormer: Learned attention uncovers tissue boundaries in spatial transcriptomics cell state classification

- Independently developed hypothesis and structured project to test if learned spatial attention over local tissue neighborhoods improved interpretability of cell-state classification in spatial transcriptomics, relative to standard non-spatial baselines.
- Designed a transformer encoding each Visium spot jointly with its 6 nearest spatial neighbors, summing expression, coordinate, and token-type embeddings so attention weights over neighbors are learned.
- Validated on 10x mouse brain data (2,796 Visium spots, 2,000 HVGs), which showed via attention analysis that the model allocates disproportionate weight (54%) to spatial neighbors at anatomical boundaries vs. 2% in homogeneous tissue, and validated gene-saliency maps against canonical cell-type markers (Mbp, Gfap, Pcp4).
- Achieved 89% validation accuracy; Logistic-regression baseline outperforms on raw accuracy (92%), but the model's learned attention provides interpretable, biologically grounded insight that the baseline cannot offer.
- Deployed for public inference on HuggingFace Spaces: huggingface.co/spaces/Varsini-Sakthi/Spatialformer

Project: Homeostatic, phagocytic, NRF2/Hmox1 oxidative, Apoc1 and chemokine microglia transcriptional programmes in naïve mouse cerebral cortex from embryo to adulthood

- Characterized microglial transcriptional heterogeneity across 5 developmental stages using pseudobulk RNA-seq pipeline integrating gene module scoring, PCA, differential gene expression, and pseudotime trajectory inference.
- Identified 4 distinct transcriptional programmes with coordinately shifting activity across development, revealing stage-specific regulatory transitions relevant to neuroinflammation and brain maturation,
- Results establish a developmental baseline for distinguishing normal microglial maturation from pathological activation in neurodevelopmental disease models.
- Writing first-author manuscript in preparation and developing tables and figures for publication.

Project: NRF2/HO-1 Antioxidant Pathway Dysregulation in Rodent Models of Perinatal Brain Injury: Rat Microglia and Mouse Oligodendrocyte Precursor Transcriptomes [Working Title]

- Analyzed bulk RNA-seq (rat microglia/macrophage) and microarray (mouse O4+ oligodendrocyte precursor cells) datasets to characterize NRF2/HO-1 antioxidant pathway dysregulation in rodent models of perinatal brain injury.
- Built domain-split protein-protein interaction networks (STRINGdb) and assessed cross-branch concordance to identify shared regulatory nodes.
- Identified that O4+ oligodendrocyte precursor cells mount a more diagnostically separable transcriptional response to IL-1β than microglia, and nominating OPCs as an additional vulnerable cell population in perinatal brain injury.

Project: Translational Validation of NRF2/HO-1 Antioxidant Signaling in Human Perinatal Brain Injury: A Cross-Cohort Transcriptomic Analysis [Working Title]

- Validated rodent-derived NRF2/HO-1 pathway findings in human perinatal brain injury by integrating transcriptomic data across 4 independent human cohorts.
- Applied timepoint-adjusted limma differential expression and pseudobulk clustering to assess translational concordance between rodent and human NRF2/HO-1 pathway signatures.

- Results showed the MHC-I/interferon/chemokine-NRF2/Hmox1 signature identified in rodents was recoverable across independent human cohorts once developmental timepoint was accounted for, extending the rodent-model findings toward clinical relevance.

Master's Thesis, *Johns Hopkins University, Department of Bioinformatics,* Baltimore, MD *Dec. 2025 – Apr. 2026*

Project: scRNA-seq Representation Learning for Cell-Type Classification

- **Developed** the hypothesis that a nonlinear VAE embedding (scVI) would outperform linear dimensionality reduction (PCA) for cell-type classification, and designed a controlled benchmarking framework to isolate the effect of the embedding method.
- Trained a 30-D scVI variational autoencoder in Python against a PCA baseline on 12,012 PBMC cells (36,601 genes, 2,000 HVGs).
- Annotated 12 immune cell populations via Leiden clustering and marker-gene differential expression (Wilcoxon test).
- Results showed PCA outperformed scVI consistently, indicating that added model complexity did not translate to better downstream performance.

Research Assistant, *Johns Hopkins University,* Baltimore, MD *Jun. 2025 – Mar. 2026*

Project: Multiclass Brain Tumor Classification from MRI Radiomics

PI: Dr. Sajung Yun

- Built a multiclass brain tumor classification pipeline on 3,000+ MRI slices.
- Achieved 97% accuracy with XGBoost; applied SHAP to show GLCM texture descriptors of tumor heterogeneity drove classification, aligning with established radiological malignancy markers.
- Reported macro-averaged precision/recall (0.59/0.63) alongside accuracy to surface reduced performance on minority tumor classes leading to the next phase of clinically urgent subtype classification.

Summer Intern, *Johns Hopkins University School of Medicine,* Baltimore, MD *May 2025 – Sep. 2025*

Project: Immune Dysregulation in a Murine Cystic Fibrosis Model

PI: Dr. Dao-Fu Dai

- Quantified immune dysregulation in a murine cystic fibrosis model via qPCR, ELISA, flow cytometry, and confocal immunofluorescence imaging of CD45+, CD3+, F4/80+, and Ly6G+ marker populations.

PUBLICATIONS

Published

- Antony A., **Sakthivadivel Ramasamy V.,** Veerappapillai S., Karuppasamy R. (2025). "Genome mining of secondary metabolites from *Streptomyces spp*. for the screening of novel and potent fungicide: Focus on rice blast disease." Research Journal of Chemistry and Environment 29(10):66–75.
- **Sakthivadivel Ramasamy V.,** Bhor S.S. (2023). "An Integrated Study to Extrapolate the Interaction of NMDAR with Potential Ligands for the Treatment of Alzheimer's Disease Symptoms." International Journal of Biochemistry and Biomolecule Research 1(1):15–27.

Manuscripts in Preparation

- **Sakthivadivel Ramasamy V**., Ozen M. "Homeostatic, phagocytic, NRF2/Hmox1 oxidative, Apoc1 and chemokine microglia transcriptional programmes in naïve mouse cerebral cortex from embryo to adulthood." Manuscript in review.
- **Sakthivadivel Ramasamy V.,** Ozen M. "NRF2/HO-1 Antioxidant Pathway Dysregulation in Rodent Models of Perinatal Brain Injury: Rat Microglia and Mouse Oligodendrocyte Precursor Transcriptomes." [Working Title]
- **Sakthivadivel Ramasamy V.,** Ozen M. "Translational Validation of NRF2/HO-1 Antioxidant Signaling in Human Perinatal Brain Injury: A Cross-Cohort Transcriptomic Analysis." [Working Title]
- **Sakthivadivel Ramasamy V.,** Ozen M. "SpatialFormer: Learned attention uncovers tissue boundaries in spatial transcriptomics cell state classification." [Working Title]

TEACHING EXPERIENCE

Teaching Assistant, Johns Hopkins University, Bioinformatics *Aug. 2025 – May 2026*

- Supported instruction of Python, UNIX, and SQL for a graduate-level bioinformatics computing course, leading lab sessions and troubleshooting student code and pipeline errors.

- Built automated RNA-seq and proteomics analysis pipelines in Python and R for in-class demonstrations and as reusable templates for student workflows, translating research-grade pipeline code into accessible teaching material.

PROFESSIONAL EXPERIENCE

Data Analyst Intern – INGEST Project, Johns Hopkins University *Oct. 2024 – Apr. 2025*

- Built a computer vision data quality assurance pipeline (CVAT, 3DSRBench, Python) to validate infant keypoint annotations for a developmental research study, applying inter-rater reliability metrics to flag discrepancies.

Project Intern – BioAxis DNA Research Centre *Jan. 2024 – Apr. 2024*

- Investigated amyloid and tau pathology in Alzheimer's disease through genomics, proteomics, and structural biology analysis, applying immunoinformatics and SNP/alternative splicing analysis to characterize proteins retrieved and cross-referenced public expression datasets (GEO) and genomic annotations (UCSC Genome Browser) to contextualize candidate genes and regulatory regions implicated in neurodegeneration.
- Analyzed drug development strategies for Transthyretin Amyloidosis, examining candidate molecular targets and therapeutic options relevant to protein misfolding disease.
- Explored vaccine optimization for *Staphylococcus aureus* using computational structural biology tools (RasMol, PVS, EMBOSS, ArgusLab), including antigenic prediction to identify candidate epitopes for vaccine design.

Intern – BioNome, Genomics & Bioinformatics Solutions *Jan. 2023 – Apr. 2023*

- Applied molecular docking (PyRx, AutoDock), MD simulations (GROMACS), QSAR modeling, and ADMET/pharmacophore screening to evaluate ligand-target binding for Alzheimer's disease drug candidates.
- Modeled and validated target protein structures using homology modeling (MODELLER) and structure-prediction tools (AlphaFold), assessing model quality with ProSA-web and ExPASy prior to docking.
- Performed SNP calling from whole-exome sequencing data using Galaxy-based NGS workflows and subtractive proteomics for hypothetical protein annotation,
- Improved identification of novel disease-relevant targets and potential drug candidates from previously uncharacterized genomic regions.
- **Wrote a first-author journal article**; prepared tables and figures, and managed the paper publication process.

SKILLS

Machine Learning: Supervised/unsupervised learning, representation learning, variational autoencoders (VAE), XGBoost, Deep learning (PyTorch), hyperparameter tuning, LLMs, model evaluation (ROC-AUC), feature engineering

Programming: Python, R, SQL, Bash, Linux/Unix, MatLab, AWS, Jupyter, Git

Bioinformatics & Genomics Tools: RNA-seq analysis, differential gene expression, single-cell RNA-seq (Scanpy, scVI), cell type annotation, transcriptomic data analysis, gene regulatory network analysis, NGS data analysis (Trimmomatic, Bowtie2/STAR), AutoDock, PyRx, GROMACS, Seurat, Discovery Studio, GEO, UCSC Genome Browser, AlphaFold, MODELLER, Galaxy, ExPASy, ProSA-web

LEADERSHIP & SERVICE

- **Vice President of Operations**, Johns Hopkins Product Management Club, 2026
- **Co-Chair of Events & Programming Committee**, AAP Student Government Association, 2025
- Project development at Kindling by PAVA center, 2025
- **Secretary**, IEEE-Electron Devices Society, 2023
- **Events Head,** Smile Over Stress, 2022